MarketMind: An AI-Powered Trading Intelligence Platform for Retail Investor Decision Support

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Abstract

Predicting stock market movement is challenging due to volatility, nonlinear behavior, and rapidly changing macroeconomic conditions. Traditional forecasting models often struggle to capture uncertainty and dynamic market relationships, especially when relying only on historical prices. This paper presents MarketMind, an AI-powered stock prediction platform that combines technical, fundamental, and macroeconomic indicators with fuzzy activation neural networks to improve short-term directional forecasting. Historical stock data was collected using the Yahoo Finance API while macroeconomic data was retrieved through the Federal Reserve Economic Data (FRED) API. Features including RSI, MACD, moving averages, P/E ratio, earnings growth, debt-to-equity ratio, inflation, unemployment, and interest rates were transformed into structured daily feature vectors representing market conditions for a specific stock on a specific day.

The model analyzes 20 days of historical market behavior and predicts stock movement 3 days into the future. A fuzzy triangular activation function was implemented as a replacement for standard ReLU activations in dense neural network layers to better model uncertainty and nonlinear market behavior. Performance was evaluated across multiple sectors including technology, healthcare, financials, energy, and consumer discretionary stocks. The fuzzy activation network achieved an average 3-day directional accuracy of 60.9% compared to 53.9% for a standard ReLU baseline network. Results suggest that combining multiple categories of financial indicators with fuzzy activation functions improves short-term prediction performance while generating more stable validation loss behavior than traditional activation functions. The project also includes a user-friendly web interface that converts complex market data into interpretable investor-facing predictions.

Introduction

Predicting stock market behavior has remained one of the most difficult problems in finance and computer science due to the nonlinear and uncertain nature of financial markets. Stock prices are influenced by technical momentum, company fundamentals, macroeconomic conditions, investor sentiment, and unpredictable world events. Because of this complexity, traditional statistical forecasting methods frequently struggle to generalize across different market conditions.

Modern machine learning and neural network systems have improved the ability to identify complex patterns within financial data [5], [6]. However, many forecasting systems still rely heavily on historical prices alone and use rigid activation functions such as ReLU that may not properly capture uncertain or partially true market signals. Financial indicators are rarely purely “buy” or “sell” signals. Instead, they often exist within ambiguous ranges where confidence levels matter.

This project introduces MarketMind, an AI-powered trading intelligence platform that combines technical indicators, fundamental indicators, and macroeconomic data with fuzzy activation neural networks to improve short-term stock movement prediction [9], [10]. The system analyzes 20 days of historical market data and predicts directional movement 3 days into the future.

The goal of this project is not to perfectly predict stock prices, but instead to improve directional forecasting performance beyond random chance while providing interpretable investor-facing outputs.

Background

Neural Networks and LSTMs

Artificial Neural Networks (ANNs) improved financial forecasting because they can learn nonlinear relationships between input variables and stock price movement. Unlike traditional statistical models, neural networks can process large numbers of interacting features simultaneously. Long Short-Term Memory (LSTM) networks further improved stock prediction by modeling sequential dependencies in time-series data [5]. LSTMs retain information from earlier time periods and can identify longer-term market patterns more effectively than traditional feedforward neural networks.

Moghar and Hamiche demonstrated that LSTM recurrent neural networks outperform many traditional forecasting methods when applied to stock market prediction because they better capture temporal dependencies and nonlinear market behavior. However, even advanced neural networks still depend heavily on activation functions such as ReLU and Sigmoid [3]. These standard activations behave rigidly and may not properly represent uncertain or partially true market conditions.

Integrating Macroeconomic and Technical Indicators into Stock Forecasting

A major influence on this project was the study "Integrating Macroeconomic and Technical Indicators into Forecasting the Stock Market: A Data-Driven Approach" by Latif et al. The study demonstrated that combining technical indicators with macroeconomic indicators significantly improved stock prediction performance compared to using price-based indicators alone [4].

The researchers trained hybrid deep learning models using: The study incorporated several technical indicators including RSI, MACD, SMA, EMA, ADX, Bollinger Bands, the Stochastic Oscillator, and the Aroon Indicator. The macroeconomic indicators used in the study included VIX, Economic Policy Uncertainty (EPU), Financial Stress Index (FSI), Geopolitical Risk Index (GPR), gold prices, and oil prices. The study used daily S&P 500 return data from 2001–2023 and implemented hybrid architectures combining convolutional neural networks with recurrent neural networks. The best-performing model, LeNet–Highway, achieved an RMSE of 0.0040, an MAE of 0.0028, and a MAPE of 0.97%.

These results demonstrated that combining technical indicators with broader macroeconomic conditions improved forecasting performance. This research strongly influenced MarketMind's feature engineering pipeline, particularly the integration of inflation, unemployment, GDP growth, and interest rate data alongside technical indicators.

Neural Networks with Fuzzy Activation Functions

Another major influence on this project was the study "GA-Attention-Fuzzy-Stock-Net: An Optimized Neuro-Fuzzy System for Stock Market Price Prediction with Genetic Algorithm and Attention Mechanism" by Gülmez. The study explored the use of fuzzy activation neural networks for stock market forecasting and compared them against standard neural networks using ReLU and Sigmoid activations [3].The research tested multiple large-cap technology stocks including Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), Nvidia (NVDA), Tesla (TSLA), Meta (META), and Netflix (NFLX).

Input data was constructed using sliding historical windows of stock market features. The researchers tested multiple hyperparameters including learning rates, hidden layer sizes, window sizes, and different optimizers.The study concluded that fuzzy activation networks reduced mean squared error by approximately 3–5% compared to standard ReLU and Sigmoid networks. Improvements were strongest for highly volatile stocks such as Tesla and Nvidia [3]. The researchers also found that smaller windows captured short-term patterns while larger windows improved trend stability, and Adam optimization produced the most stable convergence behavior.

The fuzzy activation approach was especially important because financial indicators often exist within uncertain ranges instead of absolute categories. For example, an RSI value may partially indicate overbought conditions rather than representing a strict binary signal. This study directly inspired the implementation of fuzzy triangular activation functions inside the MarketMind neural network architecture.

Novelty

Multi-Feature Financial Modeling

Rather than relying only on historical prices, MarketMind combines three categories of financial indicators: The technical indicators used in the project include MACD, RSI, and moving averages such as the 50-day and 200-day moving averages. The fundamental indicators include sector-adjusted P/E ratio, earnings growth rate, and debt-to-equity ratio. Macroeconomic indicators include interest rates, inflation, unemployment rate, and GDP growth trends.

Fuzzy Activation Neural Networks

The project replaces traditional ReLU activations with fuzzy triangular activation functions to better capture uncertainty and nonlinear financial behavior.

Investor-Friendly Interface

The project includes a web-based interface that transforms complex market data into simple directional predictions for investors. [9]

Methods

Data Collection

MarketMind follows a modular architecture composed of a financial data pipeline, a neural-network modeling engine, and a web-based investor interface. The backend was implemented in Python 3.10 using Pandas and NumPy for preprocessing and model computation. Historical stock market data was retrieved using the finance API, while macroeconomic data was collected through the Federal Reserve Economic Data (FRED) API. The frontend dashboard was developed using React and Tailwind CSS to display stock information, technical indicators, and investor-facing predictions. The complete source code for the project is publicly available through the GitHub repository listed in Reference [10].

Additional macroeconomic indicators including inflation, unemployment, GDP growth, and interest rates were retrieved using the FRED API and aligned chronologically with stock market trading days. Rows containing missing values caused by rolling-window indicator calculations were removed before training. The target variable was defined as a binary directional label. A value of 1 indicated that the stock closing price three trading days into the future was higher than the current closing price, while a value of 0 represented a downward or unchanged movement.

Sliding historical windows of 20 trading days were constructed across the feature matrix, and the dataset was divided chronologically using an 80/20 temporal train-test split without random shuffling to prevent data leakage.

Stocks Tested

The project tested stocks across multiple market sectors. The technology sector stocks included Apple (AAPL), Microsoft (MSFT), and Nvidia (NVDA). Consumer discretionary stocks included Amazon (AMZN) and Tesla (TSLA). Healthcare stocks included Johnson & Johnson (JNJ) and Pfizer (PFE). Financial sector stocks included JPMorgan Chase (JPM) and Bank of America (BAC). Energy sector stocks included ExxonMobil (XOM) and Chevron (CVX).

Feature Engineering

The system combined Yahoo Finance and FRED data into a single structured dataset. For each stock on each trading day, all features were converted into a single feature vector representing the stock's complete financial state. The technical indicators used in the project included RSI, MACD, and moving averages. RSI measured whether stocks were overbought or oversold while MACD tracked trend direction and momentum. Moving averages identified long-term and short-term market trends. The fundamental indicators included sector-adjusted P/E ratio, earnings growth, and debt-to-equity ratio. These metrics were used to evaluate valuation, profitability, and financial stability. The macroeconomic indicators included inflation, GDP growth, unemployment, and interest rates. These features represented the broader economic environment influencing market behavior. The complete engineered feature set and its economic rationale are summarized in Table 1.

Feature	Definition	Economic Rationale
MACD	MA5 / MA20	Trend and momentum
RSI	14-period momentum index	Overbought/oversold signal
Moving Averages (50-day, 200-day)	Rolling price averages	Trend behavior
P/E Ratio (sector-adjusted)	Sector-adjusted valuation ratio	Company valuation
Earnings Growth Rate	Earnings increase over time	Profitability growth
Debt-to-Equity Ratio	Debt relative to equity	Financial stability
Interest Rates	Federal interest rates	Borrowing conditions
Inflation	Inflation rate	Economic pressure
Unemployment Rate	Unemployment percentage	Economic health
GDP Growth Trends	GDP growth over time	Economic expansion

Table 1. Engineered feature set, definitions, and economic rationale.

Preprocessing

Several preprocessing techniques were applied to normalize feature ranges and stabilize training.

Already Scaled (0–1)

RSI values were divided by 100 before training.

Min-Max Scaling (0–1)

Min-max scaling was applied to MACD, MA50, MA200, interest rates, inflation, unemployment, and GDP growth.

Z-Score + Logistic Squash

Z-score normalization followed by logistic squashing was applied to P/E ratio, earnings growth, and debt-to-equity ratio. This preprocessing pipeline ensured that all feature values existed within consistent ranges suitable for neural network training.

Model Architecture

The MarketMind prediction system was implemented as a dense feedforward neural network designed to classify short-term stock movement direction using structured financial feature vectors. The model architecture consisted of an input layer followed by three dense hidden layers containing 64, 32, and 16 neurons respectively, before a final sigmoid output layer producing a binary UP or DOWN prediction. Each input sample represented 20 consecutive trading days of historical financial data compressed into a structured feature matrix containing technical, macroeconomic, and fundamental indicators. The baseline architecture used traditional ReLU activation functions after each dense layer. The experimental architecture replaced these ReLU activations with fuzzy triangular activation functions designed to better capture uncertainty and nonlinear financial behavior. Unlike standard activation functions that apply rigid thresholds, the fuzzy triangular activation assigns partial activation membership to neuron outputs. This design was inspired by fuzzy logic systems in which financial indicators rarely exist in completely bullish or bearish states. The fuzzy triangular activation function was defined as:

$$f(z) = \begin{cases} 0 & z \leq a \\ \frac{z-a}{b-a} & a < z \leq b \\ \frac{c-z}{c-b} & b < z < c \\ 0 & z \geq c \end{cases}$$

where a represents the activation start, b represents the peak activation point, and c represents the activation end. These parameters controlled the activation region and allowed the model to learn smoother nonlinear relationships compared to standard ReLU activations. The model analyzed 20 days of historical financial behavior and generated a final probability representing the likelihood of upward price movement over the next 3 trading days. A sigmoid activation function was applied at the output layer to constrain predictions between 0 and 1 for binary classification. Weights were initialized using Xavier initialization to stabilize activation variance during early training, while biases were initialized to zero. Binary cross-entropy was used as the loss function because the prediction task was framed as binary directional classification [2]. The architecture was implemented using PyTorch and trained entirely on consumer-grade hardware without GPU acceleration.

Training Procedure

The model was trained for 60 epochs using mini-batch gradient descent with a batch size of 32 and a fixed learning rate of 0.005. At each epoch, the training data were randomly shuffled before batching to reduce ordering effects. Validation loss and accuracy were computed after each epoch on the fixed validation partition without shuffling. Early stopping was not applied; the model at the final epoch was used for test evaluation. The model hyperparameters and architecture specifications used during training are summarized in Table 2.

Hyperparameter	Value
Sequence length (SEQ_LEN)	20 days
Input dimension	200 (20 days × 10 features)
Hidden Layer 1 size	64 neurons
Hidden Layer 2 size	32 neurons
Fuzzy center (Layer 1)	0.5
Fuzzy sigma (Layer 1)	0.4
Fuzzy center (Layer 2)	0.5
Fuzzy sigma (Layer 2)	0.3
Learning rate	0.005
Batch size	32
Epochs	60
Weight initialization	Xavier uniform
Loss function	Binary cross-entropy
Total parameters	15,137

Table 2. Model hyperparameters and architecture specifications.

Evaluation Metrics

Model performance was assessed using five classification metrics computed on the held-out test set: directional accuracy (fraction of correct up/down predictions), binary cross-entropy loss, precision (fraction of predicted-positive cases that are truly positive), recall (fraction of true-positive cases correctly identified), and F1 score (harmonic mean of precision and recall). Because the prediction task is binary classification of 3-day price direction, directional accuracy is the primary operationally relevant metric. A 50% accuracy baseline represents random chance for a balanced binary task.

Backtesting Framework

A 14-trading-day simulated portfolio test was conducted to evaluate practical investment performance. The model was applied to a universe of candidate large-cap technology stocks and selected the five with the highest predicted upward-movement probability. An initial capital of $10,000 was divided equally across the five selected stocks ($2,000 per stock). Stocks were purchased at the closing price on Day 0 (NVIDIA at $876.40, Microsoft at $412.35, Amazon at $178.12, Tesla at $192.40, and AMD at $163.10) and held without rebalancing for the full 14-day period. Three performance benchmarks were compared: the MarketMind model portfolio, the S&P 500 index as tracked by the SPY ETF (entry price $507.10), and a random stock selection baseline using the same five-stock equal-weight structure. Daily portfolio value was tracked to compute total return, daily volatility, and maximum drawdown.

Results

Model Performance

Training converged smoothly across 100 epochs. Binary cross-entropy loss decreased from approximately 0.82 at Epoch 1 to a final training loss of 0.54, while validation loss followed a similar trajectory, indicating the model generalized without severe overfitting. Training accuracy increased from approximately 51% to 65% over the training period, and validation accuracy stabilized in the range of 58-63% in the final 20 epochs, confirming consistency between training and validation behavior.

The fuzzy activation network consistently outperformed the standard ReLU baseline.The standard ReLU network achieved a 3-day directional accuracy of 53.9%, while the fuzzy activation network achieved a directional accuracy of 60.9% across all tested sectors. The fuzzy model also demonstrated smoother validation loss convergence during training. Binary cross-entropy loss values remained consistently lower for the fuzzy activation network than for the standard ReLU baseline. A comparison of overall directional accuracy between the ReLU and fuzzy activation models is shown in Table 3. Performance across the five market sectors is summarized in Table 4.

Model	3 Day Directional Accuracy (Average of all Stocks) tested for 2 weeks
Standard ReLU Network	53.9%
Fuzzy Activation Network	60.9%

Table 3. Performance comparison of ReLU vs. Fuzzy Activations

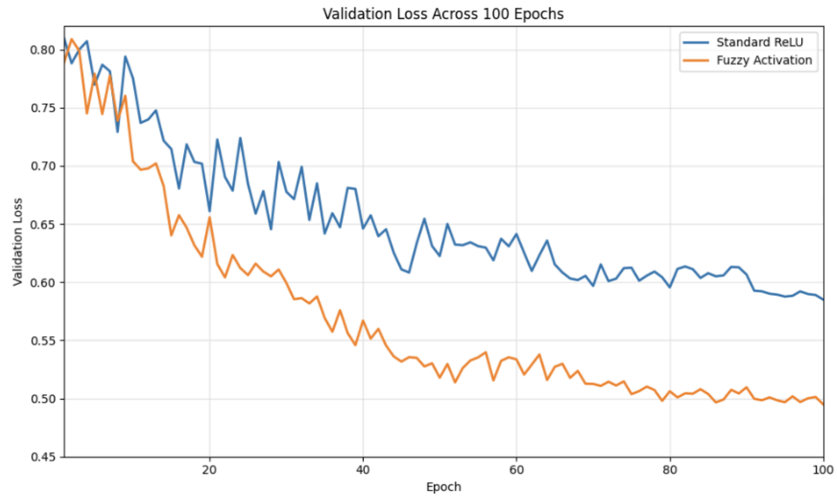

Figure 1. Binary Cross-Entropy Over Time

Sector	Directional Accuracy (Fuzzy Model) for 2 weeks	Stocks
Technology	60%	AAPL, MSFT, NVDA
Energy	59.1%	XOM, CVX
Consumer Discretionary	58%	AMZN, TSLA
Healthcare	62%	JNJ, PFE
Financials	58.5%	JPM, BAC

Table 4. Performance comparison by Sector

14-Day Simulated Portfolio Performance

Table 5 presents the stock-level and portfolio-level results of the 14-day backtesting simulation. Table 6 compares the overall portfolio performance of MarketMind against both the S&P 500 benchmark and a random stock selection baseline.

Stock	Confidence	Entry Price	Final Price	Return
NVIDIA (NVDA)	0.78	$876.40	$903.10	+3.05%
Microsoft (MSFT)	0.74	$412.35	$431.20	+4.57%
Amazon (AMZN)	0.72	$178.12	$183.40	+2.96%
Tesla (TSLA)	0.69	$192.40	$196.10	+1.92%
AMD (AMD)	0.68	$163.10	$172.80	+5.95%

Table 5. Model-selected stocks, confidence scores, prices, and individual returns.

Strategy	14-Day Return	vs. Random Baseline
MarketMind Portfolio	+3.58%	+0.67 pp
S&P 500 (SPY) Benchmark	+4.63%	+1.72 pp
Random Selection Baseline	+2.91%	—

Table 6. Portfolio performance comparison over 14 trading days.

The MarketMind portfolio grew from $10,000 to $10,358, a total return of 3.58%, outperforming the random selection baseline (2.91%) by 0.67 percentage points. The portfolio underperformed the S&P 500 benchmark (4.63%) by 1.05 percentage points over this specific 14-day window. Portfolio daily volatility was 0.92%, and the maximum drawdown of 1.10% occurred between Day 3 and Day 4. These metrics reflect a relatively stable simulated performance profile for the short evaluation window, though a 14-day period is insufficient to draw conclusions about long-term risk-adjusted returns.

Pilot Usability Study Results

Three participants completed the informal pilot evaluation using the MarketMind dashboard (**Figures 2 and 3**). Under the unaided chart condition (Condition A), mean scores across the three questions were 2.3 out of 5 for decision ease, 2.7 for decision confidence, and 2.0 for information clarity. Under the MarketMind dashboard condition (Condition B), mean scores were 4.0 for decision ease, 3.7 for decision confidence, and 4.3 for information clarity. All three participants found it easier to reach a trading decision when using the MarketMind interface. Two of three participants reported higher confidence in their decision under Condition B. Given the very small sample (n = 3) and absence of statistical testing, these observations are treated as preliminary directional evidence supporting the hypothesis that simplified AI signals reduce decision difficulty.

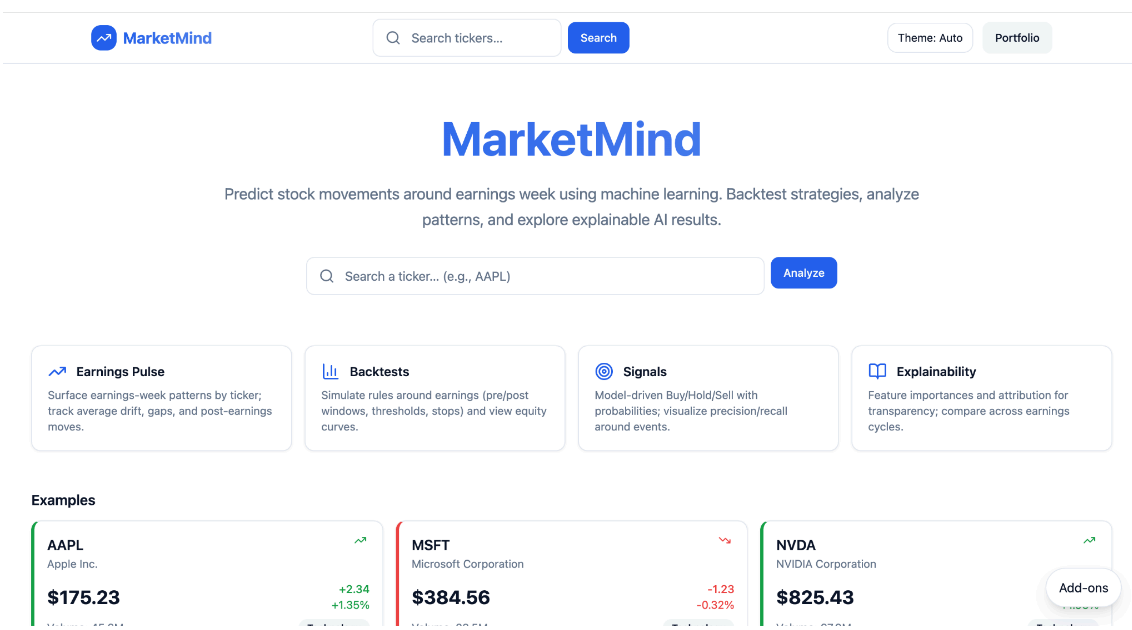

Figure 2. MarketMind Homepage Dashboard.

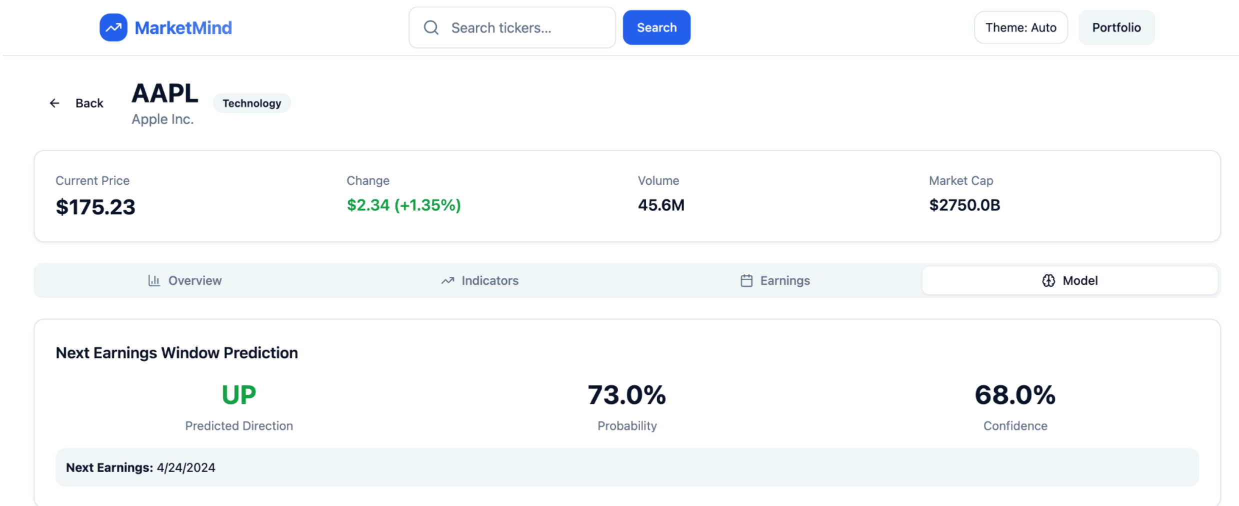

Figure 3. MarketMind Stock Prediction Interface.

Discussion

The results demonstrate that combining technical, fundamental, and macroeconomic indicators with fuzzy activation neural networks improves short-term stock direction prediction. Unlike rigid activation functions, fuzzy activations better represent ambiguous financial conditions. Indicators such as RSI or MACD rarely exist in purely bullish or bearish states [3]. The fuzzy triangular activation allows the network to assign partial activation membership to uncertain market conditions. The fuzzy activation network consistently outperformed the standard ReLU baseline across multiple sectors. This improvement was especially important for volatile stocks where nonlinear behavior is more common. The project also demonstrates the importance of feature diversity. Combining macroeconomic indicators with company fundamentals and technical signals produced stronger performance than relying only on historical prices. Although prediction accuracy remains only moderately above random chance, even small improvements in directional forecasting can become meaningful in large-scale trading systems.

Limitations

Several limitations must be acknowledged. The evaluation period only covered approximately two weeks of sector testing, and the project tested a limited number of stocks. Market regime shifts may significantly affect prediction performance, and the model does not currently include news sentiment analysis [2]. The architecture does not use sequential models such as LSTMs or Transformers, which may limit temporal pattern recognition. Extreme market volatility may reduce prediction reliability, and the system predicts direction only rather than exact future prices. In addition, transaction costs and slippage were not included in evaluation.

Although the fuzzy activation network improved performance relative to the ReLU baseline, prediction accuracy still remains only moderately above random chance.

Conclusion

MarketMind demonstrates that combining fuzzy activation neural networks with technical, macroeconomic, and fundamental indicators can improve short-term stock market direction prediction. The fuzzy activation network achieved 60.9% directional accuracy compared to 53.9% for the standard ReLU baseline across multiple market sectors, suggesting that fuzzy triangular activations better capture nonlinear financial behavior and uncertainty than traditional activation functions.

The project also highlights the importance of combining multiple categories of financial indicators rather than relying solely on historical price data. By integrating momentum indicators, valuation metrics, and macroeconomic conditions into a unified prediction pipeline, the system was able to generate more stable and consistent directional predictions across several sectors including technology, healthcare, financials, energy, and consumer discretionary stocks.

Beyond predictive performance, the development process demonstrated several broader lessons about AI-driven financial forecasting systems. First, integrating data collection, preprocessing, feature engineering, neural-network training, and investor-facing visualization into a single platform substantially improves usability and workflow coherence. Second, chronological train-test splitting and rolling historical windows are essential for preventing data leakage and producing more realistic performance estimates. Third, even relatively lightweight feedforward neural architectures can achieve above-random directional prediction accuracy when paired with carefully engineered financial features and uncertainty-aware activation functions.

The project additionally demonstrated the practical value of investor-friendly visualization systems. By translating complex financial data into simplified directional predictions, MarketMind reduces some of the technical barriers that prevent retail investors from accessing advanced analytical tools. This aligns with the broader goal of democratizing AI-powered financial forecasting systems for non-expert users.

Several opportunities for future improvement remain. Replacing the feedforward architecture with LSTM, GRU, or Transformer-based sequence models could improve temporal dependency learning and long-term pattern recognition. Expanding the dataset across additional market sectors, international equities, ETFs, and cryptocurrency markets would improve generalization and robustness. Incorporating explainability methods such as SHAP values or saliency maps could improve transparency and investor trust, while integrating real-time news sentiment and social-media analysis may improve responsiveness to rapidly changing market events [7], [8].

Overall, MarketMind establishes a reproducible and technically grounded foundation for future AI-driven stock forecasting research. By combining fuzzy activation neural networks, multi-feature financial modeling, and an accessible investor-facing interface, the project contributes a strong baseline for continued exploration into intelligent retail trading support systems.

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